ABSTRACT

Image Tampering Detection Using Deep Learning

Digital image tampering has become increasingly common due to advanced image editing tools, creating serious challenges in digital forensics, media authentication, and cybersecurity. This project focuses on detecting image tampering by classifying images as authentic or tampered. The methodology uses Error Level Analysis (ELA) to highlight compression inconsistencies, followed by a Convolutional Neural Network (CNN) for automatic feature extraction and binary classification. Common forgery types such as splicing and copy-move manipulation are considered. The system is deployed using AWS cloud services, specifically Amazon EC2.

The dataset used is sourced from a public Kaggle repository and consists of authentic and tampered images with corresponding labels.

Link: https://www.kaggle.com/datasets/divg07/casia-20-image-tampering-detection-dataset

Tech Stack

- Programming Language: Python
- Deep Learning Framework: TensorFlow, Keras
- Libraries: NumPy, Pillow (PIL), Scikit-learn, Matplotlib, Flask.
- Dataset Source: Kaggle (CASIA 2.0 Image Tampering Dataset)
- Development Tools: VS Code, GitHub
- Deployment Platform: Amazon Web Services (AWS EC2)
- Preprocessing Technique: Error Level Analysis (ELA)

References

1. Chakraborty S, Chatterjee K, Dey P. Detection of Image Tampering Using Deep Learning, Error Levels and Noise Residuals. *Neural Processing Letters*. 2024.
   Available from: https://doi.org/10.1007/s11063-024-11448-9

2. Hosny KM, Mortda AM, Lashin NA, Fouda MM. A New Method to Detect Splicing Image Forgery Using Convolutional Neural Network. *Applied Sciences (MDPI)*. 2023.
   Available from: https://www.mdpi.com/2076-3417/13/3/1272

3. Chan KY, Abu-Salih B, Qaddoura R, Al-Zoubi AM, Palade V, Pham DS, Del Ser J, Muhammad K**.** Deep neural networks in the cloud: Review, applications, challenges and research directions.*Neurocomputing(Elsevier)*.2023.
   Available from: https://doi.org/10.1016/j.neucom.2023.126327

Submitted by:

Devananda S

MAC24MCA-2022

Project Guide:

Prof. Minu Maria Joseph